

Chapter 6: Moments

Complete Step-by-Step Study Document

Reading-Only Practice · Unnumbered Alignments · Verbatim Question Boxes

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1 Textbook Exercises

1.1 Book Ch.6, Exercise 13

Book Ch.6, Exercise 13

A fair die is rolled twice, with outcomes X for the first roll and Y for the second roll. Find the moment generating function $M_{X+Y}(t)$ of $X + Y$ (your answer should be a function of t and can contain unsimplified finite sums).

Solution. By independence of X and Y , the moment generating function of their sum is the product of their individual moment generating functions:

$$\begin{aligned}
 M_{X+Y}(t) &= E\left(e^{t(X+Y)}\right) \\
 &= E\left(e^{tX}e^{tY}\right) \\
 &= E\left(e^{tX}\right)E\left(e^{tY}\right) \\
 &= M_X(t)M_Y(t).
 \end{aligned}$$

For a fair die, X is uniformly distributed on the discrete set $\{1, 2, 3, 4, 5, 6\}$ with probability $P(X = k) = \frac{1}{6}$ for each $k \in \{1, 2, 3, 4, 5, 6\}$. Computing $M_X(t)$ explicitly:

$$M_X(t) = E\left(e^{tX}\right)$$

$$\begin{aligned}
&= \sum_{k=1}^6 P(X = k) e^{kt} \\
&= \sum_{k=1}^6 \frac{1}{6} e^{kt} \\
&= \frac{1}{6} (e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t}) \\
&= \frac{1}{6} \sum_{k=1}^6 e^{kt}.
\end{aligned}$$

Since Y has the identical distribution, $M_Y(t) = M_X(t)$. Multiplying the two MGFs:

$$\begin{aligned}
M_{X+Y}(t) &= M_X(t) M_Y(t) \\
&= \left(\frac{1}{6} \sum_{k=1}^6 e^{kt} \right) \left(\frac{1}{6} \sum_{k=1}^6 e^{kt} \right) \\
&= \left(\frac{1}{6} \sum_{k=1}^6 e^{kt} \right)^2.
\end{aligned}$$

1.2 Book Ch.6, Exercise 14

Book Ch.6, Exercise 14

Let $U_1, U_2, \dots, U_{60}$ be i.i.d. $\text{Unif}(0, 1)$ and $X = U_1 + U_2 + \dots + U_{60}$. Find the MGF of X .

Solution. First, computing the MGF of a single uniform random variable $U_1 \sim \text{Unif}(0, 1)$ for $t \neq 0$:

$$\begin{aligned}
M_{U_1}(t) &= E(e^{tU_1}) \\
&= \int_0^1 e^{tu} f_{U_1}(u) du \\
&= \int_0^1 e^{tu} (1) du \\
&= \left[\frac{e^{tu}}{t} \right]_{u=0}^{u=1} \\
&= \frac{e^{t(1)}}{t} - \frac{e^{t(0)}}{t} \\
&= \frac{e^t - 1}{t}.
\end{aligned}$$

For $t = 0$:

$$\begin{aligned}
M_{U_1}(0) &= E(e^{0 \cdot U_1}) \\
&= E(1) \\
&= 1.
\end{aligned}$$

By independence of $U_1, U_2, \dots, U_{60}$, the MGF of the sum $X = U_1 + U_2 + \dots + U_{60}$ is the product of the individual MGFs:

$$M_X(t) = E(e^{t(U_1 + U_2 + \dots + U_{60})})$$

$$\begin{aligned}
&= \prod_{i=1}^{60} E(e^{tU_i}) \\
&= (M_{U_1}(t))^{60} \\
&= \begin{cases} \left(\frac{e^t - 1}{t}\right)^{60} = \frac{(e^t - 1)^{60}}{t^{60}}, & t \neq 0, \\ 1, & t = 0. \end{cases}
\end{aligned}$$

1.3 Book Ch.6, Exercise 20

Book Ch.6, Exercise 20

Let $X \sim \text{Pois}(\lambda)$, and let $M(t)$ be the MGF of X . The cumulant generating function is defined to be $g(t) = \log M(t)$. Expanding $g(t)$ as a Taylor series

$$g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$$

(the sum starts at $j = 1$ because $g(0) = 0$), the coefficient c_j is called the j th cumulant of X . Find the j th cumulant of X , for all $j \geq 1$.

Solution. For $X \sim \text{Pois}(\lambda)$, the MGF is $M(t) = e^{\lambda(e^t - 1)}$. Computing the cumulant generating function $g(t)$:

$$\begin{aligned}
g(t) &= \log M(t) \\
&= \log(e^{\lambda(e^t - 1)}) \\
&= \lambda(e^t - 1).
\end{aligned}$$

Expanding e^t as a Taylor series around $t = 0$:

$$\begin{aligned}
e^t &= \sum_{j=0}^{\infty} \frac{t^j}{j!} \\
&= 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots \\
e^t - 1 &= \sum_{j=1}^{\infty} \frac{t^j}{j!} \\
&= t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots
\end{aligned}$$

Substituting this series back into $g(t)$:

$$\begin{aligned}
g(t) &= \lambda \sum_{j=1}^{\infty} \frac{t^j}{j!} \\
&= \sum_{j=1}^{\infty} \frac{\lambda}{j!} t^j.
\end{aligned}$$

Matching coefficients with the definition $g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$ term by term:

$$\begin{aligned}\frac{c_j}{j!} t^j &= \frac{\lambda}{j!} t^j \\ \frac{c_j}{j!} &= \frac{\lambda}{j!} \\ c_j &= \lambda \quad \text{for all } j \geq 1.\end{aligned}$$

1.4 Book Ch.6, Exercise 21

Book Ch.6, Exercise 21

Let $X_n \sim \text{Bin}(n, p_n)$ for all $n \geq 1$, where np_n is a constant $\lambda > 0$ for all n (so $p_n = \lambda/n$). Let $X \sim \text{Pois}(\lambda)$. Show that the MGF of X_n converges to the MGF of X (this gives another way to see that the $\text{Bin}(n, p)$ distribution can be well-approximated by the $\text{Pois}(\lambda)$ when n is large, p is small, and $\lambda = np$ is moderate).

Solution. For $X_n \sim \text{Bin}(n, p_n)$, computing the MGF directly from the binomial distribution:

$$\begin{aligned}M_{X_n}(t) &= E(e^{tX_n}) \\ &= \sum_{k=0}^n \binom{n}{k} p_n^k (1 - p_n)^{n-k} e^{tk} \\ &= \sum_{k=0}^n \binom{n}{k} (p_n e^t)^k (1 - p_n)^{n-k} \\ &= (1 - p_n + p_n e^t)^n \\ &= (1 + p_n(e^t - 1))^n.\end{aligned}$$

Substituting $p_n = \frac{\lambda}{n}$ into the MGF:

$$\begin{aligned}M_{X_n}(t) &= \left(1 + \frac{\lambda}{n}(e^t - 1)\right)^n \\ &= \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n.\end{aligned}$$

Taking the limit as $n \rightarrow \infty$ using the standard compound-interest limit identity $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$ with $x = \lambda(e^t - 1)$ (which does not depend on n):

$$\begin{aligned}\lim_{n \rightarrow \infty} M_{X_n}(t) &= \lim_{n \rightarrow \infty} \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n \\ &= e^{\lambda(e^t - 1)} \\ &= M_X(t).\end{aligned}$$

2 Strategic Practice

2.1 SP 6 §2 · Problem 1

SP 6 §2 · Problem 1

Find $E(X^3)$ for $X \sim \text{Expo}(\lambda)$ using the MGF of X (see also Problem 2 in the Exponential Distribution section).

Solution. For $X \sim \text{Expo}(\lambda)$, the MGF for $t < \lambda$ is:

$$\begin{aligned}
 M(t) &= E(e^{tX}) \\
 &= \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx \\
 &= \lambda \int_0^\infty e^{-(\lambda-t)x} dx \\
 &= \lambda \left[-\frac{e^{-(\lambda-t)x}}{\lambda-t} \right]_0^\infty \\
 &= \lambda \left(0 - \left(-\frac{1}{\lambda-t} \right) \right) \\
 &= \frac{\lambda}{\lambda-t} \\
 &= \left(1 - \frac{t}{\lambda} \right)^{-1}.
 \end{aligned}$$

Method 1: Differentiation. Taking successive derivatives of $M(t)$ with respect to t :

$$\begin{aligned}
 M(t) &= \left(1 - \frac{t}{\lambda} \right)^{-1} \\
 M'(t) &= (-1) \left(1 - \frac{t}{\lambda} \right)^{-2} \left(-\frac{1}{\lambda} \right) \\
 &= \frac{1}{\lambda} \left(1 - \frac{t}{\lambda} \right)^{-2} \\
 M''(t) &= \frac{1}{\lambda} (-2) \left(1 - \frac{t}{\lambda} \right)^{-3} \left(-\frac{1}{\lambda} \right) \\
 &= \frac{2}{\lambda^2} \left(1 - \frac{t}{\lambda} \right)^{-3} \\
 M'''(t) &= \frac{2}{\lambda^2} (-3) \left(1 - \frac{t}{\lambda} \right)^{-4} \left(-\frac{1}{\lambda} \right) \\
 &= \frac{6}{\lambda^3} \left(1 - \frac{t}{\lambda} \right)^{-4}.
 \end{aligned}$$

Evaluating the third derivative at $t = 0$:

$$\begin{aligned}
 E(X^3) &= M'''(0) \\
 &= \frac{6}{\lambda^3} \left(1 - \frac{0}{\lambda} \right)^{-4}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{6}{\lambda^3} (1)^{-4} \\
&= \frac{6}{\lambda^3}.
\end{aligned}$$

Method 2: Pattern Recognition via Geometric Series. Expanding $M(t)$ as a geometric series for $|t| < \lambda$:

$$\begin{aligned}
M(t) &= \frac{1}{1 - \frac{t}{\lambda}} \\
&= \sum_{n=0}^{\infty} \left(\frac{t}{\lambda} \right)^n \\
&= \sum_{n=0}^{\infty} \frac{1}{\lambda^n} t^n \\
&= \sum_{n=0}^{\infty} \left(\frac{n!}{\lambda^n} \right) \frac{t^n}{n!}.
\end{aligned}$$

Matching coefficients with the Taylor expansion $M(t) = \sum_{n=0}^{\infty} E(X^n) \frac{t^n}{n!}$:

$$E(X^n) = \frac{n!}{\lambda^n} \quad \text{for all } n \geq 0.$$

Setting $n = 3$:

$$\begin{aligned}
E(X^3) &= \frac{3!}{\lambda^3} \\
&= \frac{3 \cdot 2 \cdot 1}{\lambda^3} \\
&= \frac{6}{\lambda^3}.
\end{aligned}$$

2.2 SP 6 §2 · Problem 2

SP 6 §2 · Problem 2

If X has MGF $M(t)$, what is the MGF of $-X$? What is the MGF of $a + bX$ where a and b are constants?

Solution. By definition of the moment generating function, substituting $-X$ for X :

$$\begin{aligned}
M_{-X}(t) &= E\left(e^{t(-X)}\right) \\
&= E\left(e^{(-t)X}\right) \\
&= M(-t).
\end{aligned}$$

For the linear transformation $a + bX$, where a and b are constants:

$$\begin{aligned}
M_{a+bX}(t) &= E\left(e^{t(a+bX)}\right) \\
&= E\left(e^{at+bXt}\right) \\
&= E\left(e^{at} e^{(bt)X}\right)
\end{aligned}$$

$$\begin{aligned}
&= e^{at} E\left(e^{(bt)X}\right) \\
&= e^{at} M(bt).
\end{aligned}$$

Thus, the MGF of $-X$ is $M(-t)$, and the MGF of $a + bX$ is $e^{at}M(bt)$.

2.3 SP 6 §2 · Problem 3

SP 6 §2 · Problem 3

Let $U_1, U_2, \dots, U_{60}$ be i.i.d. $\text{Uniform}(0, 1)$ and $X = U_1 + U_2 + \dots + U_{60}$. Find the MGF of X .

Solution. Computing the MGF from basic principles for a single uniform random variable $U_1 \sim \text{Uniform}(0, 1)$ for $t \neq 0$:

$$\begin{aligned}
M_{U_1}(t) &= E\left(e^{tU_1}\right) \\
&= \int_0^1 e^{tu} f_{U_1}(u) du \\
&= \int_0^1 e^{tu}(1) du \\
&= \left[\frac{e^{tu}}{t}\right]_{u=0}^{u=1} \\
&= \frac{e^{t(1)}}{t} - \frac{e^{t(0)}}{t} \\
&= \frac{e^t - 1}{t}.
\end{aligned}$$

At $t = 0$, by continuity or direct expectation:

$$\begin{aligned}
M_{U_1}(0) &= E\left(e^{0 \cdot U_1}\right) \\
&= E(1) \\
&= 1.
\end{aligned}$$

Since $U_1, U_2, \dots, U_{60}$ are independent and identically distributed, the MGF of $X = U_1 + U_2 + \dots + U_{60}$ is the product of their individual MGFs:

$$\begin{aligned}
M_X(t) &= E\left(e^{t(U_1+U_2+\dots+U_{60})}\right) \\
&= E\left(e^{tU_1}e^{tU_2}\dots e^{tU_{60}}\right) \\
&= E\left(e^{tU_1}\right)E\left(e^{tU_2}\right)\dots E\left(e^{tU_{60}}\right) \\
&= \prod_{i=1}^{60} M_{U_i}(t) \\
&= (M_{U_1}(t))^{60} \\
&= \begin{cases} \left(\frac{e^t - 1}{t}\right)^{60} = \frac{(e^t - 1)^{60}}{t^{60}}, & t \neq 0, \\ 1, & t = 0. \end{cases}
\end{aligned}$$

2.4 SP 6 §2 · Problem 4

SP 6 §2 · Problem 4

Let $X \sim \text{Pois}(\lambda)$, and let $M(t)$ be the MGF of X . The cumulant generating function is defined to be $g(t) = \ln M(t)$. Expanding $g(t)$ as a Taylor series

$$g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$$

(the sum starts at $j = 1$ because $g(0) = 0$), the coefficient c_j is called the j th cumulant of X . Find the j th cumulant of X , for all $j \geq 1$.

Solution. For $X \sim \text{Pois}(\lambda)$, we derive the cumulants explicitly from the MGF $M(t) = e^{\lambda(e^t-1)}$. Computing the cumulant generating function $g(t)$:

$$\begin{aligned} g(t) &= \ln M(t) \\ &= \ln \left(e^{\lambda(e^t-1)} \right) \\ &= \lambda(e^t - 1). \end{aligned}$$

Using the Taylor series expansion of the exponential function:

$$\begin{aligned} e^t &= 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots = \sum_{j=0}^{\infty} \frac{t^j}{j!} \\ e^t - 1 &= t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots = \sum_{j=1}^{\infty} \frac{t^j}{j!}. \end{aligned}$$

Multiplying by λ :

$$\begin{aligned} g(t) &= \lambda \sum_{j=1}^{\infty} \frac{t^j}{j!} \\ &= \sum_{j=1}^{\infty} \frac{\lambda}{j!} t^j. \end{aligned}$$

Equating coefficients of t^j with the cumulant generating function expansion $g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$:

$$\begin{aligned} \frac{c_j}{j!} t^j &= \frac{\lambda}{j!} t^j \\ \frac{c_j}{j!} &= \frac{\lambda}{j!} \\ c_j &= \lambda \quad \text{for all } j \geq 1. \end{aligned}$$

3 Homework

3.1 HW 6 · Problem 6

HW 6 · Problem 6

Let $X_n \sim \text{Bin}(n, p_n)$ for all $n \geq 1$, where np_n is a constant $\lambda > 0$ for all n (so $p_n = \lambda/n$). Let $X \sim \text{Pois}(\lambda)$. Show that the MGF of X_n converges to the MGF of X (this gives another way to see that the $\text{Bin}(n, p)$ distribution can be well-approximated by the $\text{Pois}(\lambda)$ when n is large, p is small, and $\lambda = np$ is moderate).

Solution. Let $X_n \sim \text{Bin}(n, p_n)$ with $p_n = \frac{\lambda}{n}$. Computing the MGF of X_n directly:

$$\begin{aligned} M_{X_n}(t) &= E\left(e^{tX_n}\right) \\ &= \sum_{k=0}^n \binom{n}{k} p_n^k (1-p_n)^{n-k} e^{tk} \\ &= \sum_{k=0}^n \binom{n}{k} (p_n e^t)^k (1-p_n)^{n-k} \\ &= (1-p_n + p_n e^t)^n \\ &= (1 + p_n(e^t - 1))^n. \end{aligned}$$

Substituting $p_n = \frac{\lambda}{n}$:

$$\begin{aligned} M_{X_n}(t) &= \left(1 + \frac{\lambda}{n}(e^t - 1)\right)^n \\ &= \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n. \end{aligned}$$

Taking the limit as $n \rightarrow \infty$ using the exponential limit identity $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$ with constant $x = \lambda(e^t - 1)$:

$$\begin{aligned} \lim_{n \rightarrow \infty} M_{X_n}(t) &= \lim_{n \rightarrow \infty} \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n \\ &= e^{\lambda(e^t - 1)} \\ &= M_X(t). \end{aligned}$$

3.2 HW 6 · Problem 7

HW 6 · Problem 7

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = e^X$. Then Y has a Log-Normal distribution (which means “log is Normal”; note that “log of a Normal” doesn’t make sense since Normals can be negative). Find the mean and variance of Y using the MGF of X , without doing any integrals. Then for $\mu = 0, \sigma = 1$, find the n th moment $E(Y^n)$ (in terms of n).

Solution. For $X \sim \mathcal{N}(\mu, \sigma^2)$, the moment generating function is:

$$M_X(t) = E\left(e^{tX}\right) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}.$$

Mean of Y : Since $Y = e^X$, the expectation of Y corresponds to evaluating $M_X(t)$ at $t = 1$:

$$\begin{aligned} E(Y) &= E(e^X) \\ &= M_X(1) \\ &= e^{\mu(1) + \frac{1}{2}\sigma^2(1)^2} \\ &= e^{\mu + \frac{1}{2}\sigma^2}. \end{aligned}$$

Second Moment of Y : Since $Y^2 = (e^X)^2 = e^{2X}$, this corresponds to evaluating $M_X(t)$ at $t = 2$:

$$\begin{aligned} E(Y^2) &= E(e^{2X}) \\ &= M_X(2) \\ &= e^{\mu(2) + \frac{1}{2}\sigma^2(2)^2} \\ &= e^{2\mu + \frac{1}{2}\sigma^2(4)} \\ &= e^{2\mu + 2\sigma^2}. \end{aligned}$$

Variance of Y : Using $\text{Var}(Y) = E(Y^2) - [E(Y)]^2$ and factoring:

$$\begin{aligned} \text{Var}(Y) &= E(Y^2) - [E(Y)]^2 \\ &= e^{2\mu + 2\sigma^2} - \left(e^{\mu + \frac{1}{2}\sigma^2}\right)^2 \\ &= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} \\ &= e^{2\mu + \sigma^2} \cdot e^{\sigma^2} - e^{2\mu + \sigma^2} \cdot 1 \\ &= e^{2\mu + \sigma^2} (e^{\sigma^2} - 1). \end{aligned}$$

n th Moment for $\mu = 0, \sigma = 1$: For standard normal parameters, $M_X(t) = e^{0 \cdot t + \frac{1}{2}(1)^2 t^2} = e^{t^2/2}$. Evaluating at $t = n$:

$$\begin{aligned} E(Y^n) &= E(e^{nX}) \\ &= M_X(n) \\ &= e^{(0)(n) + \frac{1}{2}(1)^2 n^2} \\ &= e^{n^2/2}. \end{aligned}$$

3.3 HW 9 · Problem 3

HW 9 · Problem 3

Consider independent Bernoulli trials with probability p of success for each. Let X be the number of failures incurred before getting a total of r successes.

1. Determine what happens to the distribution of $\frac{p}{1-p}X$ as $p \rightarrow 0$, using MGFs; what is the PDF of the limiting distribution, and its name and parameters if it is one we have studied? Hint: start by finding the $\text{Geom}(p)$ MGF. Then find the MGF of $\frac{p}{1-p}X$, and use the fact that if the MGFs of r.v.s Y_n converge to the MGF of a r.v. Y , then the CDFs of the Y_n converge to the CDF of Y .
2. Explain intuitively why the result of (a) makes sense.

Solution. (a) Limiting Distribution via MGFs. Let $q = 1 - p$. The random variable $X \sim \text{NBin}(r, p)$ counts the number of failures before r successes.

Step 1: MGF of $\text{Geom}(p)$. For a single geometric random variable $G \sim \text{Geom}(p)$ with probability mass function $P(G = k) = q^k p$ for $k \in \{0, 1, 2, \dots\}$:

$$\begin{aligned} M_G(t) &= E(e^{tG}) \\ &= \sum_{k=0}^{\infty} e^{tk} P(G = k) \\ &= \sum_{k=0}^{\infty} e^{tk} q^k p \\ &= p \sum_{k=0}^{\infty} (qe^t)^k \\ &= \frac{p}{1 - qe^t}, \quad \text{for } qe^t < 1. \end{aligned}$$

Step 2: MGF of $\text{NBin}(r, p)$. Since $X = G_1 + G_2 + \dots + G_r$ is the sum of r independent and identically distributed $\text{Geom}(p)$ random variables:

$$\begin{aligned} M_X(t) &= E(e^{t(G_1 + G_2 + \dots + G_r)}) \\ &= \prod_{i=1}^r M_{G_i}(t) \\ &= \left(\frac{p}{1 - qe^t} \right)^r, \quad \text{for } qe^t < 1. \end{aligned}$$

Step 3: MGF of the scaled random variable $\frac{p}{1-p}X = \frac{p}{q}X$. Substituting $\frac{p}{q}t$ for t in $M_X(t)$:

$$\begin{aligned} M_{\frac{p}{q}X}(t) &= E\left(e^{t\left(\frac{p}{q}X\right)}\right) \\ &= M_X\left(\frac{tp}{q}\right) \\ &= \left(\frac{p}{1 - qe^{tp/q}} \right)^r \\ &= \frac{p^r}{(1 - qe^{tp/q})^r}, \quad \text{for } qe^{tp/q} < 1. \end{aligned}$$

Step 4: Limit as $p \rightarrow 0$ for $r = 1$. Considering the base expression inside the r th power as $p \rightarrow 0$ (with $q = 1 - p \rightarrow 1$), we encounter a $\frac{0}{0}$ indeterminate form:

$$\lim_{p \rightarrow 0} \frac{p}{1 - qe^{tp/q}} = \lim_{p \rightarrow 0} \frac{p}{1 - (1 - p)e^{tp/(1-p)}}.$$

Applying L'Hôpital's Rule by differentiating the numerator and denominator with respect to p :

$$\begin{aligned} \frac{d}{dp}(p) &= 1 \\ \frac{d}{dp} \left[1 - (1 - p)e^{tp/(1-p)} \right] &= -(-1)e^{tp/(1-p)} - (1 - p)e^{tp/(1-p)} \frac{d}{dp} \left(\frac{tp}{1-p} \right) \\ &= e^{tp/(1-p)} - (1 - p)e^{tp/(1-p)} \left[t \frac{(1-p)(1) - p(-1)}{(1-p)^2} \right] \end{aligned}$$

$$\begin{aligned}
&= e^{tp/(1-p)} - (1-p)e^{tp/(1-p)} \left[t \frac{1-p+p}{(1-p)^2} \right] \\
&= e^{tp/(1-p)} - \frac{t}{1-p} e^{tp/(1-p)} \\
&= e^{tp/(1-p)} \left(1 - \frac{t}{1-p} \right).
\end{aligned}$$

Evaluating the limit of the derivative ratio as $p \rightarrow 0$ (so $e^{tp/(1-p)} \rightarrow e^0 = 1$ and $\frac{1}{1-p} \rightarrow 1$):

$$\begin{aligned}
\lim_{p \rightarrow 0} \frac{p}{1 - (1-p)e^{tp/(1-p)}} &= \lim_{p \rightarrow 0} \frac{1}{e^{tp/(1-p)} \left(1 - \frac{t}{1-p} \right)} \\
&= \frac{1}{(1)(1-t)} \\
&= \frac{1}{1-t}, \quad \text{for } t < 1.
\end{aligned}$$

Step 5: General r . Raising the limit to the r th power:

$$\begin{aligned}
\lim_{p \rightarrow 0} M_{\frac{p}{1-p}X}(t) &= \lim_{p \rightarrow 0} \left(\frac{p}{1 - qe^{tp/q}} \right)^r \\
&= \left(\lim_{p \rightarrow 0} \frac{p}{1 - qe^{tp/q}} \right)^r \\
&= \left(\frac{1}{1-t} \right)^r \\
&= \frac{1}{(1-t)^r}, \quad \text{for } t < 1.
\end{aligned}$$

Step 6: Identification of the Limiting Distribution. The limiting MGF $(1-t)^{-r}$ for $t < 1$ is the moment generating function of the Gamma($r, 1$) distribution. Since MGF convergence implies convergence in distribution, $\frac{p}{1-p}X \xrightarrow{d} \text{Gamma}(r, 1)$ as $p \rightarrow 0$. The probability density function of the limiting Gamma($r, 1$) distribution is:

$$\begin{aligned}
f(x) &= \frac{1}{\Gamma(r)1^r} x^{r-1} e^{-x/1} \\
&= \frac{x^{r-1} e^{-x}}{(r-1)!}, \quad \text{for } x > 0.
\end{aligned}$$

(b) Intuitive Explanation. The Gamma distribution is the continuous analogue of the Negative Binomial distribution, just as the Exponential distribution is the continuous analogue of the Geometric distribution. As $p \rightarrow 0$, individual Bernoulli trials succeed with vanishingly small probability, so a large number of trials is required to achieve r successes. Rescaling the failure count by $\frac{p}{1-p}$ maintains the expected value:

$$\begin{aligned}
E\left(\frac{p}{1-p}X\right) &= \frac{p}{1-p}E(X) \\
&= \frac{p}{1-p} \left(\frac{r(1-p)}{p} \right) \\
&= r,
\end{aligned}$$

which matches the mean of the Gamma($r, 1$) distribution ($E[Y] = r \cdot 1 = r$). In this high-frequency, low-probability limit, the discrete waiting time (measured in number of failures) transitions smoothly into a continuous waiting time modeled by the Gamma distribution.